WinNovation

EXECUTIVE MASTER PROGRAM 2026-27

PREMIER TECHNOLOGY & SKILL EDUCATION

GENERATIVE AI & PROMPT ENGINEERING

Master Large Language Models (LLMs), Transformer Architectures, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases, Diffusion Models & AI Agents through 70% hands-on model deployment and production AI pipelines.



3 MONTHS	70 : 30	15+	100%
12 WEEKS INTENSIVE	HANDS-ON LABS : THEORY	PRODUCTION GEN AI APPS	PLACEMENT SUPPORT

LLMs & Transformers • Advanced Prompting & RAG • LangChain & AI Agents • Diffusion & Multimodal • AI Governance

WINNOVATION EDUCATION PVT. LTD.
Building Careers Since 2017 | Greater Noida, Delhi NCR

Hotline: **+91 92112 28030** | Email: **admissions@winnovation.org**
Web: **www.winnovation.org** | Campus: **1157, i-Thum Galleria, Alpha 2**

Accelerate Your Career with Winnovation

Bridging the divide between speculative AI hype and enterprise-grade generative intelligence engineering.



"At Winnovation, our purpose is driven by the **future of our youth**. *Generative Artificial Intelligence is redefining global industries, cognitive automation, and creative problem solving. We established Winnovation to bridge the divide between speculative AI hype and enterprise-grade software architecture—ensuring our students master* **Large Language Models, Retrieval-Augmented Generation (RAG), multimodal diffusion, and autonomous agents** *to solve real enterprise challenges while strictly upholding responsible AI governance."*

Sarbarinder Singh

Director, Winnovation Education Pvt. Ltd.

"Win With Winnovation" | Recognized Skill Incubator

NEP 2020 & Competency-Based Education (CBE) Blueprint

Winnovation curricula strictly align with India's **National Education Policy (NEP 2020)** and **CBSE Vocational Skill Education** paradigms. We replace rote memorization with experiential problem solving using Bloom's Taxonomy.

70% Hands-On AI Labs

Building live conversational AI systems, fine-tuning LLMs with LoRA/QLoRA, and architecting RAG pipelines using Pinecone and ChromaDB.

Enterprise Responsible AI

Rigorous adherence to ethical AI principles, bias detection, hallucination guardrails, and compliance with India's **DPDP Act, 2023**.

Outcome-Driven Portfolios

Every module produces verifiable public software systems (document Q&A engines, multimodal marketing studios, autonomous agents).

CBSE Vocational Assessment Scheme (100 Marks Blueprint)

Assessment Component	Weightage	Evaluation Criteria	Frequency
Continuous Formative AI Sprints	30 Marks	Weekly prompt engineering labs, vector search setups, and agent testing	Weekly Milestones (12 Sprints)
Mid-Term LLM & RAG Hackathon	20 Marks	Live 4-hour hackathon building an end-to-end RAG knowledge assistant	End of Month 1 & Month 2
Comprehensive Production Capstone	20 Marks	Full production AI system (Enterprise RAG, Multi-Agent Copilot, Creative Studio)	Final Capstone Showcase
Certified AI Code Repository & Journal	15 Marks	GitHub repository with clean modular code, evaluation metrics & live demos	Terminal Evaluation
Technical Whiteboard & Model Viva	15 Marks	Oral defense justifying prompt architecture & guardrails before AI jury	Final SDE Jury Review

Surging Global Demand for Generative AI Engineers

Transforming enterprise automation, cognitive workflows, and creative production across global technology and Fortune 500 enterprises.

88%

Enterprises Adopting Gen AI

Companies across SaaS, FinTech, Healthcare, and E-Commerce actively recruit certified Gen AI and LLM prompt engineers.

3.8×

Higher Compensation Multiplier

Specialists who combine Python engineering with LLM prompt architecture and RAG pipelines command premium industry packages.

$1.3T

Projected Global AI Market

Unprecedented exponential expansion creating over 1.2 million new cognitive computing and AI engineering jobs globally.

Industry Salary Potential in India (AmbitionBox & Levels.fyi Benchmarks)

Entry-Level (0 – 2 Years)

₹8 – ₹16 LPA

Target Roles: Junior AI Engineer, Prompt Engineer, LLM Application Developer, AI Content Automation Specialist.

Mid-Level (3 – 6 Years)

₹16 – ₹32 LPA

Target Roles: Senior Generative AI Engineer, RAG Systems Architect, AI Solutions Consultant, ML Specialist.

Senior-Level (7+ Years)

₹32+ – ₹65+ LPA

Target Roles: Principal AI Architect, Head of Generative AI, Director of AI Engineering, Chief AI Officer.

Who Should Pursue This Program?

Computer Science & IT Graduates

B.Tech, BCA, MCA, and M.Sc students looking to master cutting-edge generative AI, transformer architectures, and modern agentic workflows.

Software Engineers & Developers

Software professionals looking to transition from traditional software development to high-paying generative AI engineering and RAG design.

Entrepreneurs & Startup Founders

Founders seeking to build AI-native SaaS products, intelligent conversational assistants, and automated cognitive workflow pipelines.

Product Managers & Analytics Leads

Technical leads wanting to master enterprise prompt engineering, LLM evaluation metrics, multimodal workflows, and AI integration.

THE ECONOMIC TIMES: "Generative AI and Prompt Engineering lead the fastest-growing technology job categories across India's tech hubs."

GARTNER: "By 2026, generative AI will be integrated into over 80% of enterprise software applications worldwide."

Roadmap to Master Generative AI

A structured competency ladder transforming learners into production-ready Generative AI architects and LLM engineers.

1

AI Foundations, Python & Neural Network Basics

Weeks 1 – 2

Master Python for AI, numerical computing with NumPy and Pandas, deep learning fundamentals, backpropagation, and pre-trained NLP pipeline execution.

2

Transformer Architectures & Attention Mechanisms

Weeks 3 – 4

Deconstruct the Transformer model, multi-head self-attention mechanics, encoder-decoder models (BERT/T5), decoder-only LLMs (GPT/LLaMA), and Byte-Pair tokenization.

3

Advanced Prompt Engineering & Cognitive Patterns

Weeks 5 – 6

Implement zero-shot, few-shot, Chain-of-Thought (CoT), Tree-of-Thoughts, directional stimulus prompting, structured JSON output extraction, and prompt evaluations.

4

Retrieval-Augmented Generation (RAG) & Vector DBs

Weeks 7 – 8

Build semantic search architectures with dense text embeddings, recursive document chunking, vector databases (ChromaDB, Pinecone, FAISS), and hybrid retrieval.

5

LangChain, LlamaIndex & Autonomous AI Agents

Weeks 9 – 10

Orchestrate multi-step LLM chains, conversational memory buffers, external tool execution (function calling), and autonomous multi-agent systems using LangGraph.

6

Multimodal Diffusion, Model Fine-Tuning & Deployment

Weeks 11 – 12

Deploy vision LLMs, Stable Diffusion text-to-image models, parameter-efficient fine-tuning (LoRA/QLoRA), FastAPI model serving, safety guardrails, and interview defense.

The CAISEN Immersive Learning Cycle

1. STUDY
Live masterclasses & 1:1 expert mentoring with 24×7 recordings.

2. ATTEMPT
Daily timed prompt engineering drills & vector embedding exercises.

3. MAKE
Build live production RAG pipelines & deploy AI agent workflows.

4. REVISE
Model evaluation benchmarks, hallucination audits & safety tests.

5. TEST
Mock whiteboard rounds before senior engineering leaders & FAANG juries.

Curriculum Blueprint: Month 1 & 2

AI foundations, neural computing, Large Language Models, and advanced prompt engineering.

Month 1: Foundations of AI, Python & Neural Networks

Weeks 1 – 4 | 40 Contact Hours

Core Theoretical & Applied Topics

- Introduction to AI & ML: History and evolution of Generative AI, rule-based vs statistical vs neural models
- Python for AI: Vectorized operations with NumPy, data manipulation in Pandas, tensor basics in PyTorch
- Neural Network Mechanics: Perceptrons, activation functions (ReLU, GELU), loss functions, backpropagation
- Natural Language Processing: Text tokenization, n-grams, TF-IDF, Word2Vec, GloVe embeddings
- Pre-trained Models: Hugging Face ecosystem, loading transformer weights, transfer learning pipelines
- Ethics, Bias & Safety: Algorithmic bias, intellectual property, hallucinations, and DPDP Act 2023 compliance

Guided Laboratory Practical & Takeaways

- **Lab 1:** Python tensor pipeline training a multi-layer neural network for text classification
- **Lab 2:** Pre-trained Hugging Face transformer pipeline for zero-shot text generation
- **Lab 3:** Algorithmic bias and toxicity audit benchmark evaluating language model outputs
- **Competency:** Python AI computation, neural network training, responsible AI governance

Milestone Capstone 1: Enterprise Document Summarization & Sentiment Intelligence Engine

A production Python system utilizing Hugging Face transformer pipelines to ingest long-form corporate reports, generate executive summaries, and perform multi-dimensional sentiment scoring.

Month 2: Large Language Models, Transformers & Prompt Engineering

Weeks 5 – 8 | 40 Contact Hours

Core Theoretical & Applied Topics

- LLM Landscape: Understanding GPT-4, LLaMA 3, Claude 3.5 Sonnet, Gemini 1.5, and open-source models
- Transformer Architecture: Self-attention mechanisms, scaled dot-product attention, multi-head attention
- Tokenization & Embeddings: Byte-Pair Encoding (BPE), token cost optimization, managing context windows
- Prompt Engineering Fundamentals: System instructions, role prompting, zero-shot and few-shot exemplars
- Advanced Prompting Patterns: Chain-of-Thought (CoT), Tree-of-Thoughts, ReAct pattern, self-consistency
- Output Structuring & Security: JSON mode, Pydantic validation, prompt injection defense, red-teaming

Guided Laboratory Practical & Takeaways

- **Lab 4:** Multi-persona prompt engineering suite generating deterministic, schema-validated JSON
- **Lab 5:** Automated Chain-of-Thought reasoning engine solving multi-step logical & math puzzles
- **Lab 6:** LLM evaluation harness benchmarking accuracy, hallucination rates, and latency
- **Competency:** Advanced prompt architecture, hallucination reduction, structured schema extraction

Milestone Capstone 2: Multi-Persona Enterprise AI Content Generator & Compliance Validator

An automated generative suite creating targeted marketing copy, technical documentation, and customer responses with automated schema validation and corporate brand voice compliance.

Curriculum Blueprint: Month 3 & Enterprise AI

RAG architectures, vector databases, autonomous agents, diffusion models, and cloud deployment.

Month 3: RAG Architectures, Autonomous Agents & Multimodal AI

Weeks 9 – 12 | 40 Contact Hours

RAG Architectures & Vector Systems

- Vector Embeddings: Dense representations with OpenAI text-embedding-3 and BGE models
- Vector Databases: Indexing and query execution in ChromaDB, Pinecone, and FAISS
- Advanced RAG Pipelines: Semantic chunking, metadata filtering, query rewriting, re-ranking
- LangChain & LlamaIndex: Modular chains, prompt templates, vector retrievers, output parsers
- Autonomous AI Agents: ReAct loops, tool execution, external API calling, memory stores
- LangGraph Workflows: Multi-agent coordination, human-in-the-loop validation, stateful graphs

Multimodal, Fine-Tuning & Deployment

- Multimodal LLMs: Image understanding with GPT-4o, audio transcription with Whisper
- Diffusion Models: Stable Diffusion text-to-image architecture, latent space manipulation
- Parameter-Efficient Fine-Tuning: PEFT, LoRA, and QLoRA for domain-specific adaptations
- Production Serving: Model APIs with FastAPI, web frontends in Streamlit / Next.js
- AI Safety & Monitoring: Hallucination guardrails (NeMo), telemetry with LangSmith
- FAANG Technical Sprints: Whiteboarding AI system design, prompt security, portfolio defense

Milestone Capstone 3: Autonomous Enterprise Knowledge Base Assistant with Hybrid RAG & Re-ranking

A cloud-native conversational AI assistant indexing hundreds of complex enterprise PDFs, performing hybrid keyword-vector retrieval with Cohere re-ranking, and providing verifiable cited answers with zero hallucinations.

Industry Standard Generative AI & Engineering Stack

Foundation Models & APIs

- OpenAI (GPT-4o, Embeddings)
- Anthropic Claude 3.5 Sonnet
- Meta LLaMA 3 & Mistral AI
- Google Gemini 1.5 Pro & Hugging Face

RAG & Orchestration Frameworks

- LangChain & LangGraph
- LlamaIndex Data Framework
- Vector DBs: Pinecone, Chroma, FAISS
- Cohere Re-ranker & Unstructured.io

Multimodal, Dev & Serving Tools

- PyTorch & Hugging Face PEFT/LoRA
- Stable Diffusion & Midjourney
- FastAPI, Streamlit, Docker & AWS
- LangSmith & Weights & Biases

Certification Alignment: Fully prepares candidates for AWS Certified AI Practitioner, Google Cloud Professional Machine Learning Engineer, and EC-Council Certified AI Security Specialist credentials.

Enterprise AI Ready

Industry-Ready Project Portfolio

Production-grade Generative AI applications and autonomous agent pipelines engineered for enterprise scale.

Project 1: Enterprise Hybrid Agentic RAG & GraphRAG Multi-Source Intelligence Engine

LlamaIndex + Neo4j + Qdrant + Cohere Rerank

Business Problem: Fortune 500 enterprise requires an audit-compliant cognitive search engine over 100,000+ SEC filings, engineering schematics, and tabular reports with verifiable citation lineage and zero hallucination.

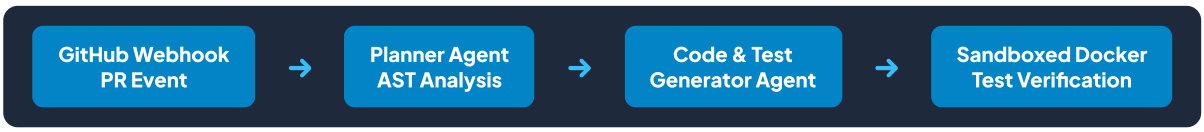


Deliverable: Production enterprise GraphRAG pipeline in FastAPI combining vector search with Neo4j entity graphs, reciprocal rank fusion (RRF), Ragas faithfulness scoring (>0.94), and sub-800ms latency.

Project 2: Autonomous Multi-Agent Software Engineering & Code Review Copilot

LangGraph + Claude 3.5 Sonnet + Docker + Tree-sitter

Business Problem: Engineering teams face code review bottlenecks and test suite regressions, requiring an autonomous multi-agent swarm to analyze pull requests, generate unit tests, and patch security vulnerabilities.



Deliverable: Multi-agent autonomous code swarm orchestrating planner, coder, and critic agents with human-in-the-loop approvals, automated AST parsing with Tree-sitter, and isolated Docker container execution.

Project 3: Enterprise Multimodal Generative Studio & Fine-Tuned LoRA Asset Engine

FLUX.1 / SDXL + ComfyUI + LoRA + GPT-4o Vision

Business Problem: Global commercial brands require on-demand photorealistic product visual generation and localized marketing ad copy while enforcing rigorous corporate typography, brand palettes, and style consistency.



Deliverable: High-throughput distributed visual generation microservice serving custom enterprise LoRA adapters, ControlNet spatial conditioning, automated safety moderation, and programmatic asset exports.

Industry-Recognized Certifications

4 official credentials validating your generative AI engineering, prompt architecture, and enterprise cloud skills to global recruiters.



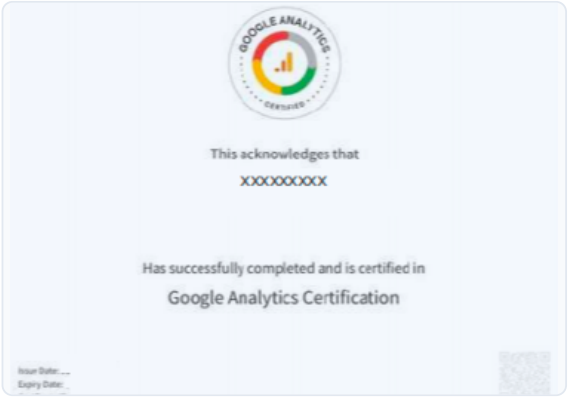
Winnovation Master Certificate in Gen AI

Awarded on successful completion of 12 weeks of training, 15+ production Gen AI applications, and jury defense of all 3 milestone capstones. Features verified QR code.



Executive AI Security & Governance Credential

ANSI-accredited international certification validating secure AI development lifecycles, prompt injection defense, model safety, and responsible AI governance.



Google Cloud Professional Machine Learning

Industry gold standard credential verifying proficiency in scalable cloud AI architectures, Vertex AI, foundation model fine-tuning, and MLOps deployment.



AWS Certified AI Practitioner Credential

Official Amazon Web Services credential validating foundational knowledge of generative AI concepts, AWS Bedrock, Amazon SageMaker, and responsible AI implementations.

Curriculum Endorsed & Co-Accredited By Global Learning Standards

Google Cloud Recognized • AWS Academy Partner • NASSCOM FutureSkills Prime • EC-Council Accredited • Open Source AI Community Aligned

Learn from Industry Experts

Seasoned AI researchers, transformer architects, and enterprise prompt engineers guiding you through every lab, pipeline, and model defense.



Abhishek Mishra

8+ Yrs Exp

Senior AI & Generative Modeling Lead

Specializes in deep transformer architectures, PyTorch fine-tuning with LoRA/QLoRA, Large Language Model alignment, and production API serving.

LLM Alignment | LoRA / QLoRA | PyTorch AI



Surekha Kanwar

10+ Yrs Exp

Senior AI Research & Neural Systems Specialist

Expert in vector embeddings mathematical modeling, semantic search optimization, high-dimensional similarity metrics, and NLP feature spaces.

Vector Search | Embeddings | Transformer NLP



Saurabh Dwivedi

5+ Yrs Exp

Senior Cloud AI & MLOps Infrastructure Lead

Architects cloud AI serving pipelines on AWS Bedrock, Pinecone vector clusters, model quantization, and low-latency API gateways.

AWS Bedrock | Pinecone Clusters | Cloud MLOps



Nisha Pal

5+ Yrs Exp

Senior AI Application & User Experience Lead

Crafts full-stack AI user interfaces in Streamlit, Next.js, FastRTC conversational interfaces, and interactive prompt evaluation portals.

LangChain UI | Streamlit Apps | AI Interfaces



Arun Kumar

5+ Yrs Exp

Senior Edge AI & Model Optimization Engineer

Focuses on Small Language Models (SLMs), model quantization with GGUF / AWQ, edge LLM deployment, and fast local inference engines.

GGUF / AWQ | Small Language Models | Edge AI



Priyanka Saini

7+ Yrs Exp

Senior Multimodal AI & Creative Systems Lead

Directs diffusion model workflows with Stable Diffusion, vision-language model integration (GPT-4o), and automated AI creative pipelines.

Diffusion Models | Vision LLMs | Multimodal AI



Pavneet Kaur

5+ Yrs Exp

Senior AI Career Coach & Interview Strategist

Prepares candidates for elite AI technical interviews, whiteboarding system design rounds, and behavioral defense before corporate juries.

System Whiteboard | STAR Defense | AI Careers



Diksha Sachan

8+ Yrs Exp

Head of Technical Curriculum & Prompt Engineering

Designs advanced prompt patterns (CoT, ReAct), LangChain autonomous agent workflows, and enterprise Retrieval-Augmented Generation labs.

Advanced Prompting | LangChain / Agents | RAG Systems



Prince Pal

5+ Yrs Exp

Enterprise AI Safety & Governance Specialist

Ensures prompt injection defense, LLM output guardrails (NeMo), algorithmic bias mitigation, and compliance with the DPDP Act 2023.

AI Guardrails | Prompt Security | DPDP Act 2023

1:1 Dedicated AI Mentorship & Live Pipeline Reviews

Weekly personalized office hours, model architecture reviews, and mock technical whiteboard interviews tailored to your target AI engineering role.

50+ Hours Live Mentorship

Student Success Stories

Hear how Winnovation graduates transitioned into high-growth Generative AI and cognitive engineering roles.



Syed Vernice Kashani

B.Tech (CSE)

Generative AI Engineer

"Mastering Generative AI and RAG architectures at Winnovation was a game-changer for my career. The hands-on labs with LangChain, Pinecone, and LLM fine-tuning gave me the exact skills needed to build production AI systems."



Prabjyot Singh

B.Tech (CSE)

Enterprise AI Solutions Associate

"I am deeply grateful to Winnovation for providing cutting-edge training in Large Language Models and vector databases. The faculty made complex concepts like attention mechanisms and LoRA fine-tuning clear and actionable."



Rabinder Singh

Govt. Technical Trainer

Senior AI & Cognitive Systems Trainer

"Winnovation's structured curriculum bridges academic concepts and production AI deployment. Mastering advanced prompt patterns and autonomous agents elevated my technical problem-solving and teaching career."



Suhaib Muzaffar

B.Tech (CSE)

LLM Application Developer

"I sincerely thank Winnovation and mentors Mr. Abhishek Mishra and Ms. Diksha Sachan for equipping me with real-world knowledge in prompt engineering and LangChain. The placement support gave me complete confidence."



Lakshay Suryavanshi

RA at NextX Tech

Generative AI Research Associate

"The practical training at Winnovation gave us direct exposure to production AI models, hybrid RAG pipelines, and diffusion systems. Building real capstone projects set my portfolio apart during hiring rounds."



Aryan Choudhary

B.Tech (CSE)

Junior AI Software Engineer

"The practical training I received in Generative AI, PyTorch, and Vector Search from Winnovation played a decisive role in shaping my career path. Their guidance and mock interviews prepared me to succeed."

100% Placement Record: Highest AI Package: ₹22 LPA | Average Starting CTC: ₹8.5 LPA Verified Alumni Network

Backed by 100+ Hiring Partners

Leading technology enterprises, cutting-edge AI startups, and global multinationals recruiting Winnovation AI engineers.

HONDA

adani Ports and Logistics

SAMSUNG

airtel

LUMINOUS Enjoy Technology

Akasa Air

SOTI

vivo

HEIZEN

SOPHOS

UTKARSH SMALL FINANCE BANK

XONIER TECHNOLOGIES

Data Resolve

DP WORLD

FORTRA

Emcure

we360.ai

ALKEM

Medtronic

arcon

motadata

FORTINET

100+

Active Corporate Recruiters

AI Product Companies, Unicorns & Global IT MNCs

15+

Interview Drives Annually

Exclusive On-Campus & Direct Hiring Drives

100%

Placement Assistance

GitHub reviews, mock whiteboards & SDE interviews

Holistic Guidance & Admissions

The complete CAISEN support framework powering your journey from aspiring learner to elite Generative AI engineer.

CAISEN 10 Core Competencies for Generative AI & Cognitive Excellence

Transformer Logic | Prompt Architecture | Vector Search | RAG Pipelines | Autonomous Agents | Model Fine-Tuning | Multimodal AI | Clean Code Architecture | AI Ethics & Safety | FAANG Whiteboarding

100% SDE Placement Support

Dedicated placement cell conducting exclusive hiring rounds with top product tech enterprises until successful career placement.

AI & GitHub Portfolio Building

ATS-compliant tech resume architecture, GitHub repository optimization, and production AI model audits by senior architects.

Rigorous Mock Whiteboards

Simulated 1-on-1 technical whiteboard rounds, live prompt defense, and behavioral STAR interviews before AI engineering leads.

Lifetime Code Repository Access

Unlimited access to high-definition recorded masterclasses, project notebooks, and continuously updated prompt template libraries.

Personal Tech Mentorship

Individual career mentoring, technical project roadmap guidance, and personal coaching to overcome technical bottlenecks.

24×7 Coding Community & TAs

Dedicated Discord and WhatsApp peer discussion channels with active teaching assistants resolving code queries within minutes.

The Winnovation Technology Ecosystem

Generative AI & Prompt Engineering | Full Stack Web Development | Data Structures & Algorithms (DSA) | Data Science & AI/ML | Data Analytics & BI | Digital Marketing & Growth | Cloud Computing & DevOps | Internet of Things (IoT) | Cyber Security & CEH

Ready to Master Generative AI & Prompt Engineering?

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Admissions Email: admissions@winnovation.org | abhishek@winnovation.org
Mobile & Helpline: +91 92112 28030 / +91 92112 28031
Campus Headquarters: 1157, i-Thum Galleria, Alpha 2, Greater Noida, Uttar Pradesh - 201308

ADMISSIONS OPEN

Batch Size: 25 Students Max

APPLY TODAY